LOOM

A cross-portfolio sourcing intelligence layer

30x the buying power. 1x the negotiating leverage.

> *"AI will redefine agility for business — making paths to growth speedy and seamless."*
>
> Think9, Principles

WHY NOW

Packaging costs are spiking this quarter

15–25%

increase in packaging input costs, 2026

1. **Crude-linked inputs**

   up 20–25% against pre-conflict levels

2. **FMCG packaging specifically**

   up 15–20%

3. **D2C margins**

   squeezed simultaneously on raw materials, packaging, logistics and manufacturing

This is margin defence against a dated, live shock — not a theoretical efficiency gain.

Sources: Packaging South Asia; Inc42, “Boxed In: The War Shock For D2C”

THE BOTTLENECK

Scale that exists on paper, not in the room

Each brand sources independently. That is the correct thing for each brand to do.

30x the buying power of any single brand

1x the negotiating leverage at the table

Because at the moment a brand negotiates, nobody in the building can see what the other brands pay for the same physical object.

Not a discipline problem. Not solved by a shared spreadsheet. And it gets monotonically worse as the portfolio grows.

Built at 11 brands. Designed for 30+.

THE HARD PROBLEM

One bottle. Five names.

`50ml Amber Glass Bottle, 20mm neck`

Neude · PDF

`Amber Boston Round 50 ML (20/400)`

Beauty by Bie · PDF

`GB-AMB-50-20N`

Panchamrit · WhatsApp

`Glass bottle - amber - 50cc - neck 20mm - w/o cap`

Neude · email

`एम्बर ग्लास बॉटल 50ml 20mm`

Goodbug · WhatsApp

ONE PHYSICAL OBJECT

You cannot combine orders for things you cannot prove are the same thing.

So the bottleneck is not price comparison. It is SKU identity resolution — and that is what a script cannot do and a model against a spec graph can.

Nobody wrote these to be inconsistent. They arrived from four vendors across PDF, email and WhatsApp.

Why not a script, and why not a hire

Could a script do it?

No.

There is no fixed input schema. ~200 vendors, each with their own quote layout, new ones onboarding constantly. Any regex-based parser is obsolete within a quarter.

Could a team do it?

Too slow.

3–4 FTE of tedious, error-prone work — and quotes expire in 15–30 days. Value exists only inside that window. A human queue introduces exactly the latency that destroys it.

Why agentic?

It is a loop.

Low-confidence extraction must trigger a clarifying action. Resolution must decide match-or-create against growing state. Detection runs continuously as quotes arrive and lapse.

Perceive → reason → act → observe, against persistent state. That is an agent, not a transform.

THE MONEY

A surcharge we pay only because we order apart

> *"Suppliers may accept 500 units if buyers agree to pay 20–30% more per unit."*
>
> Industry reporting on 2026 packaging procurement

Ordering separately

4 ×

Four brands, each below MOQ, each paying the sub-MOQ premium — four separate times.

Ordering together

0 ×

One consolidated order clears the MOQ. The premium does not apply at all.

The saving is not "we negotiated harder." It is "we stopped paying a surcharge that existed only because we ordered separately."

Arithmetic, not skill — which is why it is defensible. Negotiating leverage is upside on top.

SYSTEM ARCHITECTURE

Six layers, three human gates

L0	INGEST	Email, WhatsApp, file drop, ERP export — stored immutably with provenance	
L1	EXTRACTION	Line items with per-field confidence and source-span citation. Never infers a missing value	CONFIDENCE ROUTING
L2	RESOLUTION	Line item → canonical SKU. Attribute normalization, tolerance rules, fuzzy fallback	GATE 1
L3	SPEC GRAPH	Canonical SKUs, vendors keyed on GSTIN, brands, price history, procurement mode	
L4	OPPORTUNITY	Five always-on detectors: bundling, price outlier, concentration, lead-time drift, expiry	
L5	BRIEF	The negotiation packet a human takes into the call, with reasoning shown	
L6	DECISION	Category manager approves. Outcome writes back — the graph learns	GATES 2 & 3

The agent never commits spend and never contacts a vendor.

WHY IT MUST BE CENTRAL

Value that compounds instead of adding up

Most AI proposals

Linear

Help one team do one thing faster. Deploy to 30 brands, get 30 units of value, pay 30 units of cost.

LOOM

Superlinear

Every quote any brand receives makes the graph denser — which makes every future negotiation across all other brands better informed.

The next brand starts at portfolio-best

- Brand 12 does not negotiate up from scratch — it inherits portfolio-best pricing, qualified vendors and known lead times on day one.
- Think9 grows partly by acquisition. An acquired brand arrives with its own vendor list and its own naming conventions — canonical resolution is exactly the tool that absorbs it.
- This only works if the system sits at the centre. Which is what “centralized intelligence layer” asks for.

Three gates, placed where errors cost most

1

Canonical creation

L2

Sourcing analyst

Graph pollution. A wrong merge silently corrupts every downstream recommendation and is hard to detect afterwards.

2

Spend recommendation

L6

Category manager

Committing money on a machine judgment. The agent proposes consolidation; a human authorises it.

3

Vendor communication

L6

Category manager

Reputational and negotiating-position damage. The agent drafts; a human sends.

Confidence routing

`≥ 0.90`
auto-advance

`0.60 – 0.90`
human review

`< 0.60`
reject to human

Human effort scales with ambiguity, not with volume.

Full autonomy here would be a design error, not a feature.

PROOF OF CONCEPT

It runs. No API key, 1.8 seconds.

12
vendor artifacts

27
line items

8
canonical SKUs

5
Think9 brands

Deliberately incompatible formats

3 PDF letterheads · 3 CSVs with mutually contradictory column schemas · 2 email bodies with prices in prose · 3 WhatsApp transcripts including Hinglish and Devanagari · 1 Excel with merged cells

Real, not scaffolded

- Resolution cascade and the contradiction rule
- Five detectors computing from a live SQLite graph
- Every rupee figure derived, none asserted

The normalizer was written before the quote data existed and has not been tuned against it.

```
python run_demo.py
```

PROOF OF CONCEPT — OUTPUT

Five names resolved. One bundle. A rupee figure.

GLS-AMB-050-20

50ml amber glass bottle, 20mm neck

Neude	Beauty by Bie	Panchamrit	Goodbug
800	**600**	**450**	**700**
Rs 20.75	Rs 21.50	Rs 22.50	Rs 22.80

2,550 units of requirement vs MOQ 2,500 → clears

Rs 13,255

saved on this SKU by consolidating four brands into one order

The refusal matters as much as the match

A 48ml bottle scored 85% description similarity against the 50ml canonical — exactly the fuzzy-match threshold. A text-based matcher merges them. LOOM refuses: volume 48 ≠ 50, both stated.

Had it merged, it would have added 1,000 phantom units to the bundle above.

Rs 31,672.60

total identified across six audited findings, from 12 quotes

23 / 27 resolved (85%)
4 held for human review

Risk findings are excluded from the total — they quantify exposure against spend already counted, not new money.

IMPLEMENTATION

Thirty days to a banked saving

Scoped to packaging only — highest cross-brand overlap, substantial spend, and no regulatory sign-off to slow the loop.

Days 1–7

Ingest & extraction

Live for packaging. Seed ~150 canonical SKUs from the top 3 brands by spend.

EXIT

Extraction F1 ≥ 0.90

Days 8–15

Resolution & graph

Backfill 90 days of history via GST e-invoice records. GSTIN as vendor key.

EXIT

80% of spend resolved

Days 16–23

First real negotiation

Opportunity agent live. Run one consolidated negotiation end to end.

EXIT

One banked rupee saving

Days 24–30

Gates & measurement

Human-gate console, alerting, instrumentation. Publish the baseline.

EXIT

Cycle time, before / after

A system that banks one real saving in month one earns the right to expand to ingredients, logistics and services. One that produces a dashboard does not.

STACK

Boring on purpose

Extraction	**Claude vision, OCR fallback**	Removes the brittle OCR-to-parse chain for photographed quotes
Orchestration	**Temporal, not LangGraph**	Negotiations span days and must survive restarts. Durability beats graph ergonomics
Storage	**Postgres + pgvector, not Neo4j**	At this volume the graph is small; recursive CTEs handle the traversals
Serving	**FastAPI + Streamlit console**	The human gates must exist by day 30. Streamlit builds in days

Scale sanity check

24,000

quote line-items per year. About 100 a working day.

~$500

a year in inference cost, against a savings target in crores.

A small-data problem wearing a big-data costume.

No Kafka. No Spark. No vector database. Refusing infrastructure the scale does not justify is itself a design decision.

FAILURE MODES

What would make me wrong

Wrong canonical merge	Two different SKUs unified, and every recommendation built on a false equivalence.	Gate 1. All merges reversible with audit trail. Precision weighted above recall.
Contract manufacturers buy the packaging	If the co-packer procures, the brand isn't buying it and there is nothing to bundle.	Every brand-SKU edge carries procurement mode. Direct-buy bundles; CM-embedded gets rate-benchmarked.
Vendors learn we bundle	Base quotes inflate in anticipation of consolidated volume.	Do not disclose portfolio volume until commit. Maintain and use BATNA vendors.
Over-consolidation	Single-sourcing a bundled SKU creates a new portfolio-level supply risk.	Concentration is a first-class alert, surfaced alongside every saving.

An architecture that has not enumerated its failure modes has not been designed.